

Optimization using calculus

Our standard approaches to optimization assume that the objective function is differentiable. This allows us to use calculus to identify the solution, or at least a candidate for a solution. In my experience, there are a few common blind spots or problem areas: (1) first-order conditions are *not generally sufficient* conditions, (2) the solution *need not be interior*, and (3) second-order conditions are *more subtle in higher dimensions*. At our level, however, it is important to be rigorous. Everything must be substantiated. Thus, it is expected that you know how to address sufficiency (e.g. through correct second-order conditions) and how to account for corner solutions.

These notes focus on the blind spots, in a variety of settings. A common theme is that you can often learn a lot, and catch possible errors, by taking the time to *inspect the problem* and *reflect on your solution*. Throughout these notes, all functions are assumed to be continuously differentiable.

This material supplements MWG and adds more examples and occasionally more theoretical details, some of which are taken from Sydsæter et al (2008). Martin Osborne has a very useful online book that is both succinct and easy to navigate. It is here: <https://mjo.osborne.economics.utoronto.ca/index.php/tutorial/index/1/int/i>

1 Problems in $\mathbb{R}$

1.1 Corner solutions in $\mathbb{R}$

The following example illustrates a common mistake, specifically the possibility of corner solutions is erroneously ignored.

Example 1: Suppose you are asked to solve the problem

$$\max_{x \geq 0} tx - \frac{1}{2}(x+1)^2,$$

where $t > 0$ is some parameter. The objective function is strictly concave, so this “ought to” be easy. The necessary first-order condition (FOC) for an interior solution is

$$t - (x+1) = 0.$$

The FOC appears to have solution

$$x = t - 1,$$

so we might conclude that we are done. But, we are not!

Sanity check: Always double-check whether your solution “makes basic sense” in the context of the problem (more on this in the fourth section). In this case, the problem stipulates the restriction that $x \geq 0$. However, the candidate solution, $x = t - 1$, need not be positive. Indeed, if $t \in (0, 1)$, then the candidate solution is negative, and is therefore inadmissible. In this case, the solution is at the corner where $x = 0$. One way to see this is to note that the first derivative is negative for all $x \geq 0$, so lowering x as much as possible is always optimal. Thus, we conclude that the solution is

$$x^* = \begin{cases} 0 & \text{if } t \in (0, 1) \\ t - 1 & \text{if } t \geq 1 \end{cases},$$

which we might more informatively write as $x(t)$ or $x^*(t)$ to indicate its dependence on t .

Pet peeve: Don’t mix up “first derivative” and “the first order condition”. The first order condition is the condition that the first derivative is zero. It makes sense to say that the first derivative is negative. It makes absolutely no sense to say that the first order condition is negative.

Reminder: Even if the solution to the FOC is interior, at our level it is not sufficient to exclusively focus on the FOC. It must also be argued why the FOC is sufficient to identify the correct solution, e.g. through second-order conditions. See the next section for more on the FOC.

Exercise 1: Consider the polynomial $f(x) = a_0 + \sum_{i=1}^n a_i x^i$ with domain $X = [0, 1]$, and where $n \geq 1$ is a fixed integer and $a_0, a_1, \dots, a_n$ are parameters. Unless n is small, it is generally impossible to find stationary points in closed form. However, imagine that you know that $f(x)$ is concave. Suppose also that the only question you are interested in answering is whether the maximum is interior or at one of the corners. Explain how you can answer such a question if you know the signs of a_1 and $\sum_{i=1}^n i a_i$.

1.2 Local versus global maximum in $\mathbb{R}$

We often default to using the second derivative to determine if a stationary point is a local maximum or local minimum. However, depending on the problem, the second derivative may be much more complicated than the first derivative. You may or may not need the second derivative.

By inspecting the first derivative, you can infer when the function is increasing and when it is decreasing, which in turn allows you to identify local maxima and minima. Put another way, *the first derivative on its own contains a lot of information*. For instance, if the first derivative is first positive and then negative, then the first-order condition identifies the maximum. Note that in this particular case, the function is single-peaked or quasiconcave. In all other cases, there is more than one local maximum. You must then compare the value of the function at all the local maxima.

The following (easy) exercise illustrates the point. The solution is in the appendix.

Exercise 2: Let $f(x) = 24x^4 - 80x^3 + 81x^2 - 27x + 3$ and consider the problem

$$\max_{x \in X} f(x),$$

where X is some interval. It turns out that the first derivative can, after some simplification, be written as

$$f'(x) = 96 \left(x - \frac{1}{4}\right) \left(x - \frac{3}{4}\right) \left(x - \frac{3}{2}\right).$$

Assume first that $X = [\frac{1}{4}, \frac{3}{2}]$. By inspecting $f'(x)$, identify the local maximum. Do you agree that the function is quasiconcave on X ? What are the local minima? Assume next that $X = [0, 1]$. By inspecting $f'(x)$ identify the local maxima and minima. Identify the global maximum and minimum. Do you agree that the function is quasiconcave on X ?

2 Problems in $\mathbb{R}^2$

2.1 Concavity and second-order conditions in $\mathbb{R}^2$

Consider an objective function, $f(x, y)$, defined on $\mathbb{R}^2$. The first-order conditions correctly identify a solution if f is concave in (x, y) jointly. The point of this section is to clarify that it is not sufficient that f is concave in x and y separately. That is, $f_{xx} \leq 0$ and $f_{yy} \leq 0$ are not sufficient for the first-order conditions to identify a maximum.

You should know from math camp that the correct condition for concavity is that the Hessian matrix (the matrix of second derivatives, which includes not only f_{xx} and f_{yy} but also f_{xy}) needs to be negative semi-definite [the relevant material is in Appendices M.C, M.D, and M.J of MWG]. That is, we need

$$f_{xx} \leq 0, f_{yy} \leq 0, \text{ and } f_{xx}f_{yy} - (f_{xy})^2 \geq 0.$$

It cannot be overstated how important the second part is, yet it is surprisingly often overlooked. Remember it. The next example helps build an intuitive understanding of why the second part is needed.

Example 2: Consider the objective function

$$f(x, y) = axy - x^2 - y^2,$$

where $a \in \mathbb{R}$ is a parameter. The first-order conditions are

$$\begin{aligned} ay - 2x &= 0 \\ ax - 2y &= 0. \end{aligned}$$

The first-order conditions are satisfied if $(x, y) = (0, 0)$, and uniquely so whenever $a \notin \{-2, 2\}$. Thus, this is our candidate solution. However, even though $f_{xx} = f_{yy} = -2 < 0$, the function is not concave for all values of a . In fact,

$$f_{xx}f_{yy} - (f_{xy})^2 = 4 - a^2 \text{ is non-negative only if } a \in [-2, 2].$$

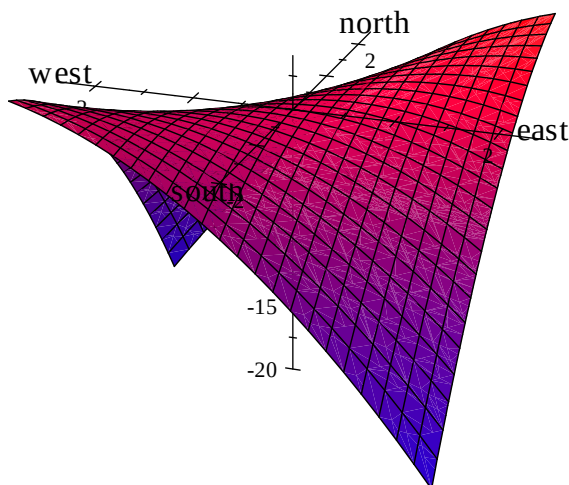
Thus, there is a problem if a falls outside $[-2, 2]$.

Here is one way to explain the problem: First, since the second derivatives are negative, it is true that if we start from $(0, 0)$ and move along the x or y axis only, then the function diminishes. To visualize, imagine that you are on what you think is the top of a hill. If you take a step east or west (change x), then you will be walking downhill. If you take a step north or south (change y), then you will also be walking downhill. However, it is too early to claim that you are at the top of the hill at $(0, 0)$ because north/south and east/west are *arbitrary* directions! How about a step to the north-east (changing both x and y)?

To see that there is a problem note that

$$\begin{aligned} f(0, 0) &= 0 \text{ the function value at the candidate solution} \\ f(z, z) &= (a - 2)z^2 \text{ when } x \text{ and } y \text{ take the same value } (z) \\ f(z, -z) &= (-a - 2)z^2 = -(a + 2)z^2 \text{ when } x \text{ and } y \text{ take opposite values.} \end{aligned}$$

When $x = y = z$ increases, it is the same as moving to the north-east. Note that if $a > 2$, then $f(z, z) = (a - 2)z^2 > 0$ for $z \neq 0$. That is, we can move *up* the hill by moving to the north-east (or the south-west). If $a < -2$, then $f(z, -z) = (-a - 2)z^2 = -(a + 2)z^2 > 0$ for $z \neq 0$ and we can move up the hill by moving to the north-west (or the south-east). Thus, we have confirmed that $(0, 0)$ is in fact not a solution when a falls outside $[-2, 2]$. Here is a plot of the function for $a = 3$:



What we learn from this example is that it is not enough that f_{xx} and f_{yy} are negative. These derivatives only capture movements along either axes. We also need to account for the possibility that we change both x and y at the same time. Indeed, in our example, the axy term means that there is an interaction between x and y , which shows up in the fact that $f_{xy} = a$ is not (generally) zero.

You do not necessarily need to know why concavity is implied precisely by $f_{xx}f_{yy} - (f_{xy})^2 \geq 0$, but the appendix contains an explanation in case you are interested.

For future reference, it turns out that cross-partial derivatives like f_{xy} play a very important role in the theory of comparative statics as well.

2.2 Notes on graphing functions or indifference curves

You will sometimes be asked to graph or sketch a function or indifference curves/level sets. I would like to clarify what the expectations are. It is generally *not* acceptable to plot the function by selecting a set of input values, calculate the corresponding function values, and then plotting and connecting the resulting points.

Whenever you have a differentiable, or even piece-wise differentiable function, the first derivative tells you what the slope is and the second derivative what the curvature is. Along with identifying any possibly points of intersection with the axes, you then have enough information to provide at least a rough sketch.

In many cases, an indifference curve can be solved for one of the variables. You can then sketch the resulting graph as explained above. The easiest possible example is when $f(x, y) = g(x) + y$, where $g(x)$ is some function. An indifference curve (at level k) is described as

$$g(x) + y = k \iff y = k - g(x),$$

which you can sketch once you know the properties of $g(x)$.

Exercise 3: In the previous example, assume that the domain is $\mathbb{R}_+^2$ and that $g(0) = 0$, $g'(x) > 0$, $g''(x) < 0$, $\lim_{x \rightarrow \infty} g(x) = t > 0$. Sketch the indifference curve.

3 Constrained optimization and the Lagrangian approach

The Lagrangian method provides an extremely impressive and powerful approach to solving constrained optimization problem. It is seemingly easy to follow the recipe and thereby solve the problem without doing much creative thinking of your own. There is a flip side:

1. If you treat the Lagrangian method as a recipe, you might lose sight of the economics. Remember, we want to solve the math problem in order to learn some economics and/or improve our intuition.

2. Do not forget to think logically about the problem at hand. The Lagrangian method can be slow. By inspecting the problem, you may be able to solve it faster, and sometimes more insightfully. For instance, you may be able to conclude that the solution is interior. Similarly, in problems with equality constraints you can substitute constraints into the objective function, thereby reducing the number of Lagrangian multipliers and choice variables.

Let us start with a brief review of the Lagrangian approach and the Kuhn-Tucker conditions. Consider some constrained maximization problem, where $\mathbf{x}$ is a vector of choice variables.¹ For simplicity, there are two constraints, an equality constraint and an inequality constraint. In addition, assume also that all variables are required to be non-negative. Hence, the problem is

$$\begin{aligned} & \max_{\mathbf{x}} f(\mathbf{x}) \\ \text{st} \quad & g(\mathbf{x}) = b \\ & h(\mathbf{x}) \leq c \\ & x_i \geq 0, \quad i = 1, 2, \dots, n. \end{aligned}$$

Technically, all the functions are supposed to be continuously differentiable on an open, convex set. We now formulate the Lagrangian,

$$\mathcal{L} = f(\mathbf{x}) - \lambda_g [g(\mathbf{x}) - b] - \lambda_h [h(\mathbf{x}) - c].$$

Note that the inequality was formulated as an $\leq$ inequality and that we put a minus sign in front of λ_h (and λ_g) in the Lagrangian. The reason this matters is that we can now sign λ_h . In particular, in any solution, $\lambda_h \geq 0$. Intuitively, from the Envelope Theorem, λ_h is the shadow price of the inequality constraint. It reveals how much better off we are if c increases. An increase in c expands the feasible set, and can therefore only be a good thing. Hence, $\lambda_h \geq 0$.² In contrast, we cannot sign λ_g . The reason is that a change in b also changes the feasible set, but it is unclear if it does so in a beneficial or detrimental manner.

We could have added the nonnegativity constraints directly to the Lagrangian. However, it is more common to deal with them in a different way, specifically by adjusting the optimality conditions.

Subject to some fine print that we return to later, the *necessary* Kuhn-Tucker conditions are as follows: If $\mathbf{x}$ is feasible and a maximum, then there are multipliers

¹The treatment of minimization problems is analogous. Alternatively, a minimization problem can always be converted into a maximization problem. In particular, a problem like $\min f(\mathbf{x})$ is equivalent to the problem $\max -f(\mathbf{x})$.

²Different sources sometimes write the Lagrangian differently. For instance, in Jehle and Reny's textbook, they write the Lagrangian to a utility maximization problem as $\mathcal{L} = u(\mathbf{x}) + \lambda(w - \mathbf{p} \cdot \mathbf{x})$ in the second edition but as $\mathcal{L} = u(\mathbf{x}) - \lambda(\mathbf{p} \cdot \mathbf{x} - w)$ in the third edition. Of course, the two are equivalent.

such that

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x_i} &\leq 0, \text{ with } \frac{\partial \mathcal{L}}{\partial x_i} = 0 \text{ if } x_i > 0, i = 1, 2, \dots, n \\ \lambda_h [h(\mathbf{x}) - c] &= 0 \text{ and } \lambda_h \geq 0,\end{aligned}$$

where

$$\frac{\partial \mathcal{L}}{\partial x_i} = \frac{\partial f(\mathbf{x})}{\partial x_i} - \lambda_g \frac{\partial g(\mathbf{x})}{\partial x_i} - \lambda_h \frac{\partial h(\mathbf{x})}{\partial x_i}$$

and where the condition that $\lambda_h [h(\mathbf{x}) - c] = 0$ and $\lambda_h \geq 0$ is often referred to as the complementary slackness condition. Thus, if $\lambda_h > 0$ then it is necessarily the case that the inequality constraint is binding, or $h(\mathbf{x}) = c$. If the constraint is slack, $h(\mathbf{x}) < c$, the multiplier must be zero, $\lambda_h = 0$. It is sometimes overlooked that it is also possible that both $\lambda_h = 0$ and $h(\mathbf{x}) = c$ at the same time.

Hence, the typical approach is to identify all $(\mathbf{x}, \lambda_g, \lambda_h)$ that satisfy the Kuhn-Tucker conditions plus the constraints in the maximization problem. There are often several solutions. Among these candidate solutions, the true maximum can be found by checking which has the highest $f(\mathbf{x})$ value.

Since there are often several candidate solutions, the typical approach can involve a lot of work. *A shortcut would be nice.* This is where *inspecting the problem* may again prove useful. A few different kinds of shortcuts are demonstrated in the following.

First, it is often possible to see right away that the solution must be interior, which in turn means that we do not have to waste time looking for corner solutions. On the other hand, it is also important not to jump to the conclusion that the solution is interior – you must provide an argument or proof in support.

Example 3: Consider a utility maximization problem where $u(x_1, x_2) = x_1^3 x_2^3$. Prices and income are strictly positive. Let λ denote the multiplier to the budget constraint. The Kuhn-Tucker conditions and the constraints are

$$\begin{aligned}3x_1^2 x_2^3 - \lambda p_1 &\leq 0 \text{ with equality if } x_1 = 0 \\ 3x_1^3 x_2^2 - \lambda p_2 &\leq 0 \text{ with equality if } x_2 = 0 \\ \lambda (p_1 x_1 + p_2 x_2 - w) &= 0 \text{ and } \lambda \geq 0 \\ p_1 x_1 + p_2 x_2 - w &\leq 0 \text{ and } x_1, x_2 \geq 0\end{aligned}$$

To find all the solutions, we have to distinguish between all cases. There are different ways of doing so. For instance, we can distinguish between $\lambda = 0$ and $\lambda > 0$, or we can distinguish between $\min\{x_1, x_2\} = 0$ and $\min\{x_1, x_2\} > 0$.

Case 1, $\lambda = 0$: To satisfy the first two conditions, at least one of x_1 or x_2 must be zero. There are infinitely many such solutions. However, all these solutions give utility zero.

Case 2, $\lambda > 0$: The budget constraint must bind, by complementary slackness. All the conditions are satisfied if all income is spent on one good and nothing on the other.

However, this means that utility is zero. It remains to look for solutions where both x_1 and x_2 are strictly positive. In this case, all the $\leq$ become $=$. The corresponding solution has strictly positive utility since $x_1, x_2 > 0$ and it must therefore be the true maximum of the problem.

In this example, there are infinitely many irrelevant solutions. However, a shortcut starts by noting that utility is zero if one or both of x_1 and x_2 are zero. This cannot be optimal, since a better bundle is affordable (one with strictly positive x_1 and x_2). Hence, the true solution must be interior. It follows from the first two conditions that $\lambda > 0$ and therefore that the budget constraint is binding. This gives us the same solution as above.

Second, in most utility maximization problems, it is clear that the budget constraint is binding. However, be careful to note that this does not imply that the solution is interior, as it is possible that the consumer purchases nothing of a subset of goods. Moreover, the curvature properties of the objective function may provide information that allows you to narrow down the set of candidate solutions.

Example 4: Consider a utility maximization problem with $u(x_1, x_2) = x_1^2 + x_2^2$. By inspecting the problem, we note that (i) preferences are locally non-satiated and (ii) the objective function is convex. The first property implies that the budget constraint binds. The second property implies that the objective function is quasiconvex, so the indifference curves bend the opposite way compared to normal. It follows that the solution will be at an extreme point where the budget constraint binds (draw a figure, or recall the exercise from the handout on convexity). In other words, all income is spent on one good. Since the objective function is symmetric in x_1 and x_2 , the solution must be to spend all income on the cheapest good.

Exercise 4: Solve the above problem using the Lagrangian approach from scratch, pretending that you did not know the shortcuts. Explain why one of the candidate solutions is interior. Is this the true solution and why/why not?

Exercise 5: Consider the utility function $u(x_1, x_2) = (2x_1 + x_2)^{\frac{1}{2}} x_1$. By inspecting the problem, what can you say about x_1 ? By inspecting the problem, what can you say about the budget constraint? Now solve the problem. The solution is in the appendix.

It is of interest to ask when the Kuhn-Tucker conditions are sufficient. In particular, any candidate solution that satisfies the sufficient conditions is a maximum, so we do not need to compare them directly. There are various forms of sufficient conditions. The one presented here relies on quasiconcavity and quasiconvexity. These are exactly the properties that appear in many economics problems. Moreover, as we discussed last week, the structure of the solution set under these assumptions is particularly nice. The following result is related to Theorem M.K.3 in MWG.

The result allows for m inequality constraints of the form $h_k(\mathbf{x}) \leq c_k$, $k = 1, \dots, m$. However, equality constraints are ruled out. Assume the domain of all functions is $\mathbb{R}^n$, (or an open subset thereof). Note that nonnegativity constraints were not mentioned, but they can be accommodated by adding in more inequality constraints. We can now state the result. A proof is in the Appendix.

Theorem: Assume there are no equality constraints. Assume that the objective function $f(\mathbf{x})$ is quasiconcave and that all $h_k(\mathbf{x})$ are quasiconvex, $k = 1, \dots, m$. Then, if $\mathbf{x}^*$ satisfies the Kuhn-Tucker conditions and $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$, then $\mathbf{x}^*$ solves the problem.

Remark: Given quasiconvexity of $h_k(\mathbf{x})$, $k = 1, \dots, m$, the constraint set, C , is a convex set. Given the objective function is quasiconcave, we then know from last week that the set of solutions is a convex set. Equality constraints are different, because they do not imply that the constraint set is convex.

Remark: If there is a solution with $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$, then it is not necessary to examine those $\mathbf{x}$ for which $\nabla f(\mathbf{x}) = \mathbf{0}$. However, if there is no such $\mathbf{x}^*$, then it does become necessary to examine the points where $\nabla f(\mathbf{x}) = \mathbf{0}$. Moreover, even if $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$, the theorem does not guarantee that $\mathbf{x}^*$ satisfy the Kuhn-Tucker conditions. More on this in the next section.

Exercise 6: Compare the utility functions $u(x_1, x_2) = (x_1 x_2)^3$ and $v(x_1, x_2) = x_1 x_2$. Assume there is a standard budget constraint and the usual nonnegativity constraints. Using the Kuhn-Tucker necessary conditions, compare the number of candidate solutions under $u(x_1, x_2)$ and $v(x_1, x_2)$. Noting that the two utility functions represent the same preferences, explain the difference between the number of candidate solutions and how it fits with the theorem.

4 The fine print: Constraint qualification

In the previous section, we skipped the fine print under which the Kuhn-Tucker conditions are necessary. The fine print is as follows:

If $\mathbf{x}$ is feasible and a maximum **and the constraint qualification is satisfied** at $\mathbf{x}$, then the Kuhn-Tucker conditions are satisfied.

As the name suggests, the constraint qualification requires that the constraints in the maximization problem satisfy certain properties. There are various versions of the constraint qualification. Here are two:

1. If all constraints (i.e. $g(\mathbf{x})$ and $h(\mathbf{x})$ above) are linear in $\mathbf{x}$, then no further conditions are required. The constraint qualification is automatic. **An implication is that the constraint qualification is not an issue in the standard utility maximization problem.**

2. The gradients of the constraints that are binding at $\mathbf{x}$ are linearly independent. This is the version mentioned in MWG.

To spell out the second case, consider again the case studied above with constraints described by $g(\mathbf{x}) = b$ and $h(\mathbf{x}) \leq c$, but ignoring nonnegativity constraints for simplicity. Assume that both constraints bind at $\mathbf{x}$, in which case it is required that $\nabla g(\mathbf{x})$ and $\nabla h(\mathbf{x})$ are linearly independent. That is, it is *not* the case that there exists scalars α and β , not both zero, for which

$$\alpha \nabla g(\mathbf{x}) + \beta \nabla h(\mathbf{x}) = \mathbf{0}.$$

To understand why there is a problem, assume that $\nabla g(\mathbf{x})$ and $\nabla h(\mathbf{x})$ are linearly dependent, with $\alpha \neq 0$. Then,

$$\nabla g(\mathbf{x}) = -\frac{\beta}{\alpha} \nabla h(\mathbf{x}).$$

The Kuhn-Tucker conditions and the binding constraints require

$$\begin{aligned} \nabla f(\mathbf{x}) - \lambda_g \nabla g(\mathbf{x}) - \lambda_h \nabla h(\mathbf{x}) &= \mathbf{0} \\ g(\mathbf{x}) &= b \\ h(\mathbf{x}) &= c. \end{aligned}$$

Assuming $\mathbf{x} \in \mathbb{R}^n$, there are a total of $n+2$ equations with $n+2$ variables, $(\mathbf{x}, \lambda_g, \lambda_h)$. However, due to linear dependency, the first n equations can be written as

$$\begin{aligned} \nabla f(\mathbf{x}) - \lambda_g \nabla g(\mathbf{x}) - \lambda_h \nabla h(\mathbf{x}) &= \mathbf{0} \\ \nabla f(\mathbf{x}) - \lambda_g \left(-\frac{\beta}{\alpha} \nabla h(\mathbf{x}) \right) - \lambda_h \nabla h(\mathbf{x}) &= \mathbf{0} \\ \nabla f(\mathbf{x}) - \left(\lambda_g \frac{\beta}{\alpha} + \lambda_h \right) \nabla h(\mathbf{x}) &= \mathbf{0}. \end{aligned}$$

Thus, while we still have $n+2$ equations, we in reality only have $n+1$ variables, namely $\mathbf{x}$ and the single number $\lambda_g \frac{\beta}{\alpha} + \lambda_h$. Thus, there is generally speaking not enough degrees of freedom to satisfy the Kuhn-Tucker conditions and the binding constraints. The implication is that if $\mathbf{x}$ is a maximum, but the constraint qualification is not satisfied at $\mathbf{x}$, then the standard recipe may *miss* $\mathbf{x}$ as a solution. Consequently, it is necessary to examine the points at which the constraint qualification is violated in isolation.

Example 5: Consider the objective function $f(x_1, x_2) = (x_1 x_2)^3$ defined on $\mathbb{R}^2$. Assume that the sole constraint is the inequality constraint that $h(\mathbf{x}) = (x_1 + x_2 - 2)^2 \leq 0$. By inspecting the constraint, we observe that it is satisfied only if $x_1 + x_2 - 2 = 0$, but if this is the case then the gradient is

$$\nabla h(\mathbf{x}) = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

which is linearly dependent. Thus, we might expect a problem. The Kuhn-Tucker conditions are

$$\begin{aligned} 3x_1^2x_2^3 - \lambda 2(x_1 + x_2 - 2) &= 0 \\ 3x_1^3x_2^2 - \lambda 2(x_1 + x_2 - 2) &= 0 \\ \lambda(x_1 + x_2 - 2)^2 &= 0 \text{ and } \lambda \geq 0. \end{aligned}$$

However, as noted, $x_1 + x_2 - 2 = 0$. Hence, the first two conditions are satisfied if and only if one or both of x_1 and x_2 are zero. They cannot both be zero, because this violates the constraint. If one is zero, the constraint dictates that that other must be 2. Hence $(2, 0)$ and $(0, 2)$ are the only points that solve Kuhn-Tucker conditions. However, the corresponding value of the objective function is zero, which is clearly not optimal. After all, the point $(1, 1)$ is feasible and better.

What is missing is to separately check all the $\mathbf{x}$ for which the constraint qualification is violated. In this particular example, it is violated on the entire feasible set. However, it is easy to solve our problem, since it is equivalent to $\max_{x_1, x_2} (x_1 x_2)^3$ subject to $x_1 + x_2 = 2$. This is a standard utility maximization problem with Cobb-Douglas preferences, $\mathbf{p} = (1, 1)$, and $w = 2$. The solution is $(1, 1)$.

Note that $f(x_1, x_2)$ is quasiconcave on $\mathbb{R}_+^2$ and satisfies $\nabla f(1, 1) \neq \mathbf{0}$. At the same time, $(x_1 + x_2 - 2)^2$ is convex and therefore quasiconvex. Hence, by the theorem in Section 4, we expect the Kuhn-Tucker conditions to be sufficient at $\mathbf{x}^* = (1, 1)$. However, the conditions are not satisfied at $\mathbf{x}^*$, due to the constraint qualification. Further, note that $\nabla f(2, 0) = \nabla f(0, 2) = \mathbf{0}$.

Reference

Sydsæter, K., Hammond, P., Seierstad, A. and A. Strom, 2008, *Further Mathematics for Economic Analysis*, 2nd edition, Pearson.

Appendix

An explanation and interpretation of $f_{xx}f_{yy} - (f_{xy})^2 \geq 0$

Take some objective function $f(x, y)$. Move ε steps to the north and $c\varepsilon$ steps to the east (if $c < 0$ then we are moving towards the west). As we change c , we change the direction that we are walking in. If we are really at the top of the hill, then there must be no direction – or no c – such that we walk up the hill. Now, as we walk, the

objective changes to $f(x + \varepsilon, y + c\varepsilon)$. For any given direction, c , the first derivative with respect to ε is

$$f_x(x + \varepsilon, y + c\varepsilon) + f_y(x + \varepsilon, y + c\varepsilon)c \rightarrow f_x(x, y) + f_y(x, y)c \text{ as } \varepsilon \rightarrow 0.$$

Hence, if the FOCs are satisfied, $f_x = f_y = 0$, then the function is flat (unchanged) if we take a tiny step in any direction. But we don't know if this is a minimum or maximum. Therefore, we need to look at the second derivative too, which is

$$\begin{aligned} f_{xx}(x + \varepsilon, y + c\varepsilon) + f_{xy}(x + \varepsilon, y + c\varepsilon)c + f_{xy}(x + \varepsilon, y + c\varepsilon)c + f_{yy}(x + \varepsilon, y + c\varepsilon)c^2 \\ \rightarrow f_{xx}(x, y) + 2f_{xy}(x, y)c + f_{yy}(x, y)c^2 \text{ as } \varepsilon \rightarrow 0. \end{aligned}$$

Now, we need to make sure that our (x, y) solution is a maximum regardless of what c is – so we need the function to be concave for *all* c ! Otherwise, we can find a direction where the function is convex, and we will start climbing up the hill.

Now note that $s(c) = f_{xx}(x, y) + 2f_{xy}(x, y)c + f_{yy}(x, y)c^2$ is (weakly) negative at $c = 0$ if $f_{xx} \leq 0$. So, we need $f_{xx} \leq 0$, or the function would be convex as we move to the north. By reversing the roles of x and y , we see that we also need $f_{yy} \leq 0$. If $f_{yy} = 0$, then the function becomes $f_{xx}(x, y) + 2f_{xy}(x, y)c$. If f_{xy} is not zero, then we can just pick a large enough c of opposite sign to f_{xy} , in which case $f_{xx}(x, y) + 2f_{xy}(x, y)c$ is made positive. In that case, we have found a direction in which the function is convex. Thus, if $f_{yy} = 0$ (or $f_{xx} = 0$) then $f_{xy} = 0$ is needed for concavity.

To continue, assume that $f_{xx}, f_{yy} \leq 0$ and allow $f_{xy} \neq 0$. Note that $s(c) = f_{xx}(x, y) + 2f_{xy}(x, y)c + f_{yy}(x, y)c^2$ is quadratic in c . In fact, since $f_{yy} \leq 0$, the function is concave in c . Thus, let us try to maximize it with respect to c (a nice and simple maximization problem in one variable) Remember that we are hoping that $s(c)$ is always negative, but this is guaranteed if the highest possible value of $s(c)$ is negative. The FOC is

$$2f_{xy}(x, y) + 2f_{yy}(x, y)c = 0.$$

If $f_{yy} = 0$, the FOC requires that $f_{xy} = 0$ as well, which is consistent with our previous discussion. In this case, $s(c)$ is a constant, and it is weakly negative if and only if $f_{xx} \leq 0$. Note that in this case, $f_{xx}f_{yy} - (f_{xy})^2 \geq 0$ is trivially satisfied. In the remainder, assume that $f_{yy} < 0$. Then, the FOC is satisfied when

$$c = -\frac{f_{xy}(x, y)}{f_{yy}(x, y)}$$

Thus, the highest value that $s(c)$ takes is

$$\begin{aligned}
\max s(c) &= f_{xx} + 2f_{xy} \left(-\frac{f_{xy}}{f_{yy}} \right) + f_{yy} \left(-\frac{f_{xy}}{f_{yy}} \right)^2 \\
&= f_{xx} - 2\frac{(f_{xy})^2}{f_{yy}} + \frac{(f_{xy})^2}{f_{yy}} \\
&= f_{xx} - \frac{(f_{xy})^2}{f_{yy}} \\
&= \frac{1}{f_{yy}} [f_{xx}f_{yy} - (f_{xy})^2].
\end{aligned}$$

We want this to be always weakly negative, to ensure concavity, but since $f_{yy} < 0$, this means that we need

$$f_{xx}f_{yy} - (f_{xy})^2 \geq 0.$$

In conclusion, the FOCs correctly identify a local maximum if the following second order conditions are satisfied:

$$f_{xx} \leq 0, f_{yy} \leq 0, \text{ and } f_{xx}f_{yy} - (f_{xy})^2 \geq 0.$$

The last condition ensures that regardless of which arbitrary direction we walk in, we will always be walking down the hill when starting from the point identified by the FOCs.

Solution to the Exercise 2

Assume first that $X = [\frac{1}{4}, \frac{3}{2}]$. As x increases from $\frac{1}{4}$ to $\frac{3}{2}$, we see that $f'(x)$ is positive until it reaches $\frac{3}{4}$ and negative thereafter. In other words, $f(x)$ is first-increasing-then-decreasing. That is, it is quasiconcave, and the stationary point at $x = \frac{3}{4}$ is the global maximum. There are local minima at the corners, i.e. when $x = \frac{1}{4}$ and $x = \frac{3}{2}$.

Assume next that $X = [0, 1]$. We see that $f'(x)$ is first negative, then positive, then negative. Hence, the first stationary point, at $x = \frac{1}{4}$, is a local minimum and the second, at $x = \frac{3}{4}$, a local maximum. However, there are also the corners. Given the shape of the function $x = 0$ is a local maximum, while $x = 1$ is a local minimum.

To identify the global maximum we compare the local maxima, observing that

$$\begin{aligned}
f\left(\frac{3}{4}\right) &= \frac{69}{32} = 2.1563 \\
f(0) &= 3.
\end{aligned}$$

Hence, $x = 0$ is the global maximum. Turning to the global minimum, we compare

$$\begin{aligned}
f\left(\frac{1}{4}\right) &= \frac{5}{32} = 0.15625 \\
f(1) &= 1,
\end{aligned}$$

and conclude that $x = \frac{1}{4}$ is the global minimum.

The function is not quasiconcave on $X = [0, 1]$. The reason is that there is a portion of the domain where the function is first decreasing and then increasing.

Solution to Exercise 5

By inspecting the problem, it is clear that no solution can feature $x_1 = 0$. The reason is that utility is zero, but the consumer can afford to earn strictly positive utility by buying a small quantity of x_1 . Hence, any solution must have $x_1 > 0$, but it remains the case that $x_2 = 0$ is possible.³

We can also see that utility is locally non-satiated. The reason is that a small increase in x_1 always increases utility. It follows that the budget constraint is binding.

To solve the problem, the Lagrangian is

$$\mathcal{L} = (2x_1 + x_2)^{\frac{1}{2}} x_1 - \lambda [p_1 x_1 + p_2 x_2 - w],$$

with

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_1} &= (2x_1 + x_2)^{-\frac{1}{2}} x_1 + (2x_1 + x_2)^{\frac{1}{2}} - \lambda p_1 \leq 0 \\ \frac{\partial \mathcal{L}}{\partial x_2} &= \frac{1}{2} (2x_1 + x_2)^{-\frac{1}{2}} x_1 - \lambda p_2 \leq 0. \end{aligned}$$

Note that if $\lambda = 0$ then the second condition implies that $x_1 = 0$, but in this case utility must be zero, which cannot be optimal. Hence $\lambda > 0$, which means that the budget constraint is binding. So, we learned this from the mathematical approach, but in a way this obscures the economics. The economics is that preferences are locally non-satiated and so the budget constraint must bind; we already knew this just by looking at the problem.

Since $x_1 > 0$, the first Kuhn-Tucker condition must bind. It remains to figure out if $x_2 = 0$ or $x_2 > 0$. We can utilize the fact that the budget constraint binds.

Case 1, $x_2 = 0$. We can use the budget constraint to solve for $x_1 = \frac{w}{p_1}$ and then substitute this and $x_2 = 0$ into the $\frac{\partial \mathcal{L}}{\partial x_1} = 0$ condition to derive the value of λ ,

$$\begin{aligned} \lambda &= \frac{1}{p_1} \left[(2x_1)^{-\frac{1}{2}} x_1 + (2x_1)^{\frac{1}{2}} \right] \text{ since } x_2 = 0 \\ &= \frac{1}{p_1} \left[(2)^{-\frac{1}{2}} + (2)^{\frac{1}{2}} \right] x_1^{\frac{1}{2}} \\ &= \frac{1}{p_1} \left[(2)^{-\frac{1}{2}} + (2)^{\frac{1}{2}} \right] \left(\frac{w}{p_1} \right)^{\frac{1}{2}}. \end{aligned}$$

³On a rather technical note, the objective function $(2x_1 + x_2)^{\frac{1}{2}} x_1$ is defined only when $2x_1 + x_2 \geq 0$. However, $2x_1 + x_2 = 0$ is not optimal. Thus, we can restrict attention to $2x_1 + x_2 > 0$. The set of (x_1, x_2) that satisfy this condition is an open set. On this open set, the objective function is continuously differentiable, as technically required by the Kuhn-Tucker conditions.

Let us now check if this is consistent with the second Kuhn-Tucker condition that $\frac{\partial \mathcal{L}}{\partial x_2} \leq 0$,

$$\begin{aligned}\frac{1}{2} (2x_1)^{-\frac{1}{2}} x_1 - \frac{p_2}{p_1} \left[(2)^{-\frac{1}{2}} + (2)^{\frac{1}{2}} \right] x_1^{\frac{1}{2}} &\leq 0 \\ \frac{1}{2} (2)^{-\frac{1}{2}} - \frac{p_1}{p_1} \left[(2)^{-\frac{1}{2}} + (2)^{\frac{1}{2}} \right] &\leq 0\end{aligned}$$

and multiply by $2(2)^{\frac{1}{2}} p_1$ to get

$$\begin{aligned}p_1 - p_2 [2 + 2 \times 2] &\leq 0 \\ p_1 - 6p_2 &\leq 0,\end{aligned}$$

so in order for $x_2 = 0$, it is necessary that $p_1 - 6p_2 \leq 0$. Otherwise, i.e. when $p_1 - 6p_2 > 0$, it must be the case that $x_2 > 0$.

Case 2, $x_2 > 0$. In this case, the solution is interior and $\frac{\partial \mathcal{L}}{\partial x_1} = \frac{\partial \mathcal{L}}{\partial x_2} = 0$. Since $\lambda > 0$, rearranging and dividing the two yields the standard tangency condition. We already know the budget constraint is binding. Thus, we have two equations with two unknowns, x_1 and x_2 . You can verify that the solution is

$$\begin{aligned}x_1 &= \frac{2}{3} \frac{w}{p_1 - 2p_2} \\ x_2 &= \frac{1}{3} \frac{w}{p_2} \frac{p_1 - 6p_2}{p_1 - 2p_2}.\end{aligned}$$

Note that this solution says that $x_1 > 0$ if and only if $p_1 - 2p_2 > 0$. But we already know that $x_1 > 0$ must be part of the solution, so there is definitely a problem with the above if $p_1 - 2p_2 < 0$. In short, we cannot have $x_2 > 0$ in this case. So, $x_2 = 0$ and we are back at the boundary solution where $x_1 = \frac{w}{p_1}$.

For $x_2 > 0$ it must hold that $p_1 - 6p_2$ and $p_1 - 2p_2$ have the same sign. This holds if $p_1 - 6p_2 > 0$ or $p_1 - 2p_2 < 0$, but not otherwise. We have already dealt with the $p_1 - 2p_2 < 0$ case. This leaves the case where $p_1 - 6p_2 > 0$. However, in this case, the above solution is interior with $x_1, x_2 > 0$.

Luckily, this fits with our conclusion from case 1 (where $x_2 = 0$). There, we concluded that $p_1 - 6p_2 \leq 0$ is necessary for $x_2 = 0$. But now we have actually established that it is sufficient as well because the solution to case 2 (where $x_2 > 0$) does not make sense if $p_1 - 6p_2 < 0$.

In conclusion, the solution to the problem is

$$\begin{aligned}x_1(p, w) &= \begin{cases} \frac{2}{3} \frac{w}{p_1 - 2p_2} & \text{if } p_1 - 6p_2 > 0 \\ \frac{w}{p_1} & \text{if } p_1 - 6p_2 \leq 0 \end{cases} \\ x_2(p, w) &= \begin{cases} \frac{1}{3} \frac{w}{p_2} \frac{p_1 - 6p_2}{p_1 - 2p_2} & \text{if } p_1 - 6p_2 > 0 \\ 0 & \text{if } p_1 - 6p_2 \leq 0 \end{cases}.\end{aligned}$$

Now, if you had just solved for an interior solution, you would of course have noticed a problem (reaching the conclusion that either $x_1 < 0$ or $x_2 < 0$) if $p_1 - 6p_2 < 0$. This contradiction then implies that the solution must be at the boundary, and as we discussed earlier, you can conclude that $x_1 > 0$. So, you would have been able to find the correct Marshallian demands that way too. An alternative approach would have been to substitute the binding budget constraint into the objective function (which kills one degree of freedom) and then maximized the resulting one-variable objective function.

Proof of the theorem in Section 4

Preliminaries: First, we need to utilize another way of characterizing quasiconcavity under differentiability. This is described in Theorem M.C.3 (see Figure M.C.5 for a graphical illustration). In particular, the rule is

$$f(\mathbf{x}) \text{ is quasiconcave} \iff \nabla f(\mathbf{x}) \cdot (\mathbf{x}' - \mathbf{x}) \geq 0 \text{ whenever } f(\mathbf{x}') \geq f(\mathbf{x}).$$

Second, letting $\lambda_k \geq 0$ denote the multiplier to the k 'th constraint, note that

$$\frac{\partial \mathcal{L}(\mathbf{x})}{\partial x_i} = \frac{\partial f(\mathbf{x})}{\partial x_i} - \sum_{k=1}^m \lambda_k \frac{\partial h_k(\mathbf{x})}{\partial x_i}, \quad i = 1, 2, \dots, n,$$

or in vector notation

$$\nabla \mathcal{L} = \nabla f(\mathbf{x}) - \sum_{k=1}^m \lambda_k \nabla h_k(\mathbf{x}), \quad i = 1, 2, \dots, n.$$

Finally, we recall that the Kuhn-Tucker conditions imply that $\nabla \mathcal{L}(\mathbf{x}^*) = \mathbf{0}$ at any solution $\mathbf{x}^*$, or

$$\nabla f(\mathbf{x}^*) = \sum_{k=1}^m \lambda_k \nabla h_k(\mathbf{x}^*).$$

Proof of the Theorem: Let $\mathbf{x}^* \in C$ denote a point that satisfies the Kuhn-Tucker conditions. We want to prove that $\mathbf{x}^*$ is optimal. By contradiction, assume that there is some $\mathbf{x} \in C$ with $\mathbf{x} \neq \mathbf{x}^*$ and for which $f(\mathbf{x}) > f(\mathbf{x}^*)$.

If constraint k is binding, then $\lambda_k (h_k(\mathbf{x}) - c_k) \leq \lambda_k (h_k(\mathbf{x}^*) - c_k) = 0$, or $\lambda_k h_k(\mathbf{x}) \leq \lambda_k h_k(\mathbf{x}^*)$. If constraint k is not binding, then $\lambda_k = 0$, and it once again holds that $\lambda_k h_k(\mathbf{x}) \leq \lambda_k h_k(\mathbf{x}^*)$. Since $\lambda_k \geq 0$ and $h_k(\mathbf{x})$ is quasiconvex, it follows that $-\lambda_k h_k(\mathbf{x})$ is quasiconcave. Thus, since $-\lambda_k h_k(\mathbf{x}) \geq -\lambda_k h_k(\mathbf{x}^*)$, the above properties of quasiconcavity implies that

$$\begin{aligned} -\lambda_k \nabla h_k(\mathbf{x}^*) \cdot (\mathbf{x} - \mathbf{x}^*) &\geq 0, \text{ or} \\ 0 &\geq \lambda_k \nabla h_k(\mathbf{x}^*) \cdot (\mathbf{x} - \mathbf{x}^*) \end{aligned}$$

for all k . Summing over all the constraints,

$$\begin{aligned} 0 &\geq \sum_{k=1}^m \lambda_k \nabla h_k(\mathbf{x}^*) \cdot (\mathbf{x} - \mathbf{x}^*) \\ &= \nabla f(\mathbf{x}^*) \cdot (\mathbf{x} - \mathbf{x}^*) \end{aligned}$$

since

$$\nabla f(\mathbf{x}^*) = \sum_{k=1}^m \lambda_k \nabla h_k(\mathbf{x}^*)$$

from the Kuhn-Tucker conditions.

Since $f(\mathbf{x}) > f(\mathbf{x}^*)$ by assumption, it is also true by continuity that $f(\mathbf{x} - t\nabla f(\mathbf{x}^*)) > f(\mathbf{x}^*)$ when t is a small positive number. Fix any such $t > 0$. Since $f(\mathbf{x})$ is quasiconcave it then follows that

$$\begin{aligned} \nabla f(\mathbf{x}^*) \cdot (\mathbf{x} - t\nabla f(\mathbf{x}^*) - \mathbf{x}^*) &\geq 0 \\ \nabla f(\mathbf{x}^*) \cdot (\mathbf{x} - \mathbf{x}^*) - t\nabla f(\mathbf{x}^*) \cdot \nabla f(\mathbf{x}^*) &\geq 0 \end{aligned}$$

or

$$\begin{aligned} \nabla f(\mathbf{x}^*) \cdot (\mathbf{x} - \mathbf{x}^*) &\geq t\nabla f(\mathbf{x}^*) \cdot \nabla f(\mathbf{x}^*) \\ &= t \sum_{i=1}^n \left(\frac{\partial f(\mathbf{x}^*)}{\partial x_i} \right)_{\mathbf{x}=\mathbf{x}^*}^2. \end{aligned}$$

Keeping in mind that $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$, at least one of the terms in the sum is strictly positive. Hence,

$$\nabla f(\mathbf{x}^*) \cdot (\mathbf{x} - \mathbf{x}^*) > 0$$

We have now concluded that

$$0 \geq \nabla f(\mathbf{x}^*) \cdot (\mathbf{x} - \mathbf{x}^*) > 0,$$

which is a contradiction. Hence, $\mathbf{x}^*$ solves the problem, as claimed.